PHYS 209: Aspects of Modern Physics

Module 4: Qubits

Lecture 6: Quantum Decoherence

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Review

- In quantum mechanics states are represented by kets $|\psi\rangle$
  - Infinite dimensional systems: wave function $\psi(x,t)$
  - Finite dimensional systems: column matrix
- Qubits (2 dim) can be in a superposition state:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

- Two qubits system states can be obtained by tensor products

$$|\Psi\rangle_{A+B} = |\psi\rangle_A \otimes |\phi\rangle_B$$

- For entangled states like Bell states, such a decomposition is not possible.

$$|\phi^{\pm}\rangle = \frac{|0\rangle \otimes |0\rangle \pm |1\rangle \otimes |1\rangle}{\sqrt{2}} = \frac{|00\rangle \pm |11\rangle}{\sqrt{2}}$$

$$|\psi^{\pm}\rangle = \frac{|0\rangle \otimes |1\rangle \pm |1\rangle \otimes |0\rangle}{\sqrt{2}} = \frac{|01\rangle \pm |10\rangle}{\sqrt{2}}$$

- In such a case, we cannot define the wave function of the constituent systems.
  - Is there some other mathematical object that can represent such states?
  - Since all systems interact with environments around them and may get entangled with the environment, what is the proper way to describe real quantum systems?

The Density Matrix – Introduction

- A qubit state $|\psi\rangle$ can be described by a density matrix

$$\hat{\rho} = |\psi\rangle\langle\psi|$$

- For a general superposition state

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \qquad \langle\psi| = \alpha^*\langle 0| + \beta^*\langle 1|$$

  - The density matrix is

$$\hat{\rho} = |\psi\rangle\langle\psi| = [\alpha|0\rangle + \beta|1\rangle][\alpha^*\langle 0| + \beta^*\langle 1|]$$
$$= \alpha\alpha^*|0\rangle\langle 0| + \alpha\beta^*|0\rangle\langle 1| + \beta\alpha^*|1\rangle\langle 0| + \beta\beta^*|1\rangle\langle 1|$$

- We can write the density matrix in matrix representation

$$|\psi\rangle = \begin{bmatrix}\alpha\\ \beta\end{bmatrix}, \qquad \langle\psi| = [\alpha^* \quad \beta^*]$$

  - Then,

$$\hat{\rho} = \begin{bmatrix}\alpha\\ \beta\end{bmatrix}[\alpha^* \quad \beta^*] = \begin{bmatrix}\alpha\alpha^* & \alpha\beta^*\\ \beta\alpha^* & \beta\beta^*\end{bmatrix}$$

The Density Matrix – Matrix Representation

$$\hat{\rho} = \begin{bmatrix} \alpha\alpha^* & \alpha\beta^* \\ \beta\alpha^* & \beta\beta^* \end{bmatrix}$$

- Diagonal elements encode measurement probabilities:

$$\rho_{00} = |\alpha|^2 = P(|0\rangle), \qquad \rho_{11} = |\beta|^2 = P(|1\rangle)$$

- Property of Density Matrices: $\mathrm{Tr}[\rho] = 1$ Normalization

$$\mathrm{Tr}[\hat{\rho}] = \sum_i \rho_{ii} = \rho_{00} + \rho_{11} = |\alpha|^2 + |\beta|^2 = P(|0\rangle) + P(|1\rangle) = 1$$

- Off-diagonal elements encode phase relationship

$$\rho_{01} = \alpha\beta^*, \qquad \rho_{10} = \alpha^*\beta = \rho_{01}^*$$

  - These are called quantum coherences – they carry information that $|\psi\rangle$ is a superposition state, not just a classical mixture.
  - Note that $\rho_{10} = \rho_{01}^*$ implies $\hat{\rho}^\dagger = \hat{\rho}$. Hermitian

The Density Matrix – Pure & Mixed States

- Pure State: Density matrices of the following form represent pure states.

$$\hat{\rho} = |\psi_A\rangle\langle\psi_A|$$

  - Here the quantum state is $|\psi_A\rangle$.
  - For pure states, $\hat{\rho}^2 = (|\psi_A\rangle\langle\psi_A|)(|\psi_A\rangle\langle\psi_A|) = |\psi_A\rangle\langle\psi_A\rangle = \hat{\rho}$
  - Meaning, $\mathrm{Tr}[\hat{\rho}^2] = \mathrm{Tr}[\hat{\rho}] = 1$
- Mixed States: Density matrices can represent classical mixtures of the form where the state is in $|\psi_A\rangle$ with probability p and $|\psi_B\rangle$ with probability $1 - p$.

$$\hat{\rho} = p|\psi_A\rangle\langle\psi_A| + (1-p)|\psi_B\rangle\langle\psi_B|$$

  - This is a classical mixture of quantum states.
  - For mixed states, $\hat{\rho}^2 \neq \hat{\rho}$
  - Meaning $\mathrm{Tr}[\hat{\rho}^2] \neq 1$

In fact, $\mathrm{Tr}[\hat{\rho}^2] < 1$

System-Environment Interaction

- Any system S interacts with and is affected by its environment E.
  - System: general qubit $|\psi\rangle_S = \alpha|0\rangle_S + \beta|1\rangle_S$.
  - Environment: some unspecified system $|\phi\rangle_E$.
  - Full State:

$$|\Psi\rangle_{S+E} = |\psi\rangle_S \otimes |\phi\rangle_E = (\alpha|0\rangle_S + \beta|1\rangle_S) \otimes |\phi\rangle_E$$
$$= \alpha|0\rangle_S \otimes |\phi\rangle_E + \beta|1\rangle_S \otimes |\phi\rangle_E$$
$$= \begin{bmatrix} \alpha|\phi\rangle_E \\ \beta|\phi\rangle_E \end{bmatrix}$$

- Suppose that the interaction between the system and environment is of the form

$$\hat{H}_{\text{int}} = \hbar\hat{\sigma}_z \otimes \hat{B} = \hbar\hat{Z} \otimes \hat{B}$$

  - Then the time evolution operator is

$$\hat{U}(t) = e^{-i\hat{H}t/\hbar} = e^{-i(\hat{Z}\otimes\hat{B})t}$$

  - This tells you how $S + E$ evolves.

System + Environment State

$$|\Psi\rangle_{S+E} = \alpha|0\rangle_S \otimes |\phi\rangle_E + \beta|1\rangle_S \otimes |\phi\rangle_E$$
$$\hat{U}(t) = e^{-i\hat{H}t/\hbar} = e^{-i(\hat{Z}\otimes\hat{B})t}$$

$$\hat{A}|\psi\rangle = a|\psi\rangle$$
$$\hat{A}|\psi\rangle = a\hat{I}|\psi\rangle$$
$$f(\hat{A})|\psi\rangle = f(a\hat{I})|\psi\rangle$$
$$+$$
$$\hat{Z}|0\rangle = |0\rangle$$
$$\hat{Z}|1\rangle = -|1\rangle$$
$$\downarrow$$
$$e^{-i\hat{Z}t}|0\rangle = e^{-i\hat{I}t}|0\rangle$$
$$e^{-i\hat{Z}t}|1\rangle = e^{i\hat{I}t}|1\rangle$$

- Under this interaction, the total system $S + E$ evolves to

$$\begin{aligned}|\Psi(t)\rangle_{S+E} &= \hat{U}(t)|\Psi(0)\rangle_{S+E} \\ &= e^{-i(\hat{Z}\otimes\hat{B})t}(\alpha|0\rangle_S \otimes |\phi\rangle_E + \beta|1\rangle_S \otimes |\phi\rangle_E) \\ &= \alpha|0\rangle_S \otimes e^{-i\hat{B}t}|\phi\rangle_E + \beta|1\rangle_S \otimes e^{i\hat{B}t}|\phi\rangle_E \\ &= \alpha|0\rangle_S \otimes |E_+(t)\rangle + \beta|1\rangle_S \otimes |E_-(t)\rangle\end{aligned}$$

  - So, the interaction produces entanglement between system and environment qubits.
- Question: We know $S + E$ state after interaction. But we only care about the system. How do we get its state after time t?
  - Good news: Density matrices allow us to get state of S from $S + E$ using ‘partial trace’.
  - It will be easier to work in matrix representation:

$$|\Psi(t)\rangle_{S+E} = \begin{bmatrix}\alpha|E_+(t)\rangle \\ \beta|E_-(t)\rangle\end{bmatrix}$$

System + Environment Density Matrix

$$|\Psi(t)\rangle_{S+E} = \begin{bmatrix} \alpha|E_+(t)\rangle \\ \beta|E_-(t)\rangle \end{bmatrix}$$

- System + Environment Density Matrix:

$$\hat{\rho}_{S+E}(t) = |\Psi(t)\rangle_{S+E}\langle\Psi(t)|_{S+E}$$

$$= \begin{bmatrix} \alpha|E_+(t)\rangle \\ \beta|E_-(t)\rangle \end{bmatrix} \begin{bmatrix} \alpha^*\langle E_+(t)| & \beta^*\langle E_-(t)| \end{bmatrix}$$

$$= \begin{bmatrix} |\alpha|^2|E_+(t)\rangle\langle E_+(t)| & \alpha\beta^*|E_+(t)\rangle\langle E_-(t)| \\ \beta\alpha^*|E_-(t)\rangle\langle E_+(t)| & |\beta|^2|E_-(t)\rangle\langle E_-(t)| \end{bmatrix}$$

  - This is an operator valued 2×2 matrix.
  - The fully expanded matrix will be 4×4.
- To get the system's density matrix, we must trace over the environment's degrees of freedom.

Trace

- Consider a square matrix $A = \begin{bmatrix} A_{00} & A_{01} \\ A_{10} & A_{11} \end{bmatrix}$
- Trace:

$$\mathrm{Tr}[A] = A_{00} + A_{11} = \sum_i A_{ii}$$

  - If we write this matrix in bra-ket form:

$$A = A_{00}|0\rangle\langle 0| + A_{01}|0\rangle\langle 1| + A_{10}|1\rangle\langle 0| + A_{11}|1\rangle\langle 1|$$

- Equivalent definition of the trace:

$$\mathrm{Tr}[A] = \sum_i \langle e_i|A|e_i\rangle$$

  where $|e_i\rangle = \{|0\rangle, |1\rangle\}$ are the basis states.

  - To see this:

$$\mathrm{Tr}[A] = \langle 0|A|0\rangle + \langle 1|A|1\rangle = A_{00} + A_{11}$$

Partial Trace

$$\mathrm{Tr}[A] = \sum_i \langle e_i | A | e_i \rangle$$

- Consider a pure composite system $S + E$ with density matrix

$$\hat{\rho}_{S+E} = |\psi\rangle_S \langle\psi|_S \otimes |1\rangle_E \langle 1|_E$$

- To get density matrix of S, take partial trace over E:

$$\begin{aligned}\mathrm{Tr}_E[\hat{\rho}_{S+E}] &= \sum_i \left\langle e_i^E | \hat{\rho}_{S+E} | e_i^E \right\rangle = \langle 0^E | \hat{\rho}_{S+E} | 0^E \rangle + \langle 1^E | \hat{\rho}_{S+E} | 1^E \rangle \\ &= |\psi\rangle_S \langle\psi|_S \otimes \langle 0|1\rangle_E \langle 1|0\rangle_E + |\psi\rangle_S \langle\psi|_S \otimes \langle 1|1\rangle_E \langle 1|1\rangle_E \\ &= 0 + |\psi\rangle_S \langle\psi|_S \\ &= |\psi\rangle_S \langle\psi|_S = \hat{\rho}_S\end{aligned}$$

  - So, tracing over E recovers density matrix of S.

- For mixed states, tracing over E does not give something of the form $|\psi\rangle_S \langle\psi|_S$.

Tracing out Environment

$$\hat{\rho}_{S+E}(t) = \begin{bmatrix} |\alpha|^2 |E_+(t)\rangle\langle E_+(t)| & \alpha\beta^* |E_+(t)\rangle\langle E_-(t)| \\ \beta\alpha^* |E_-(t)\rangle\langle E_+(t)| & |\beta|^2 |E_-(t)\rangle\langle E_-(t)| \end{bmatrix}$$

- We can get the S density matrix by taking partial trace over E:

$$\hat{\rho}_S(t) = \mathrm{Tr}_E[\hat{\rho}_{S+E}(t)]$$

$$= \mathrm{Tr}_E \begin{bmatrix} |\alpha|^2 |E_+(t)\rangle\langle E_+(t)| & \alpha\beta^* |E_+(t)\rangle\langle E_-(t)| \\ \beta\alpha^* |E_-(t)\rangle\langle E_+(t)| & |\beta|^2 |E_-(t)\rangle\langle E_-(t)| \end{bmatrix}$$

$$= \begin{bmatrix} |\alpha|^2 \mathrm{Tr}_E(|E_+(t)\rangle\langle E_+(t)|) & \alpha\beta^* \mathrm{Tr}_E(|E_+(t)\rangle\langle E_-(t)|) \\ \beta\alpha^* \mathrm{Tr}_E(|E_-(t)\rangle\langle E_+(t)|) & |\beta|^2 \mathrm{Tr}_E(|E_-(t)\rangle\langle E_-(t)|) \end{bmatrix}$$

System Density Matrix

$$\hat{\rho}_S(t) = \begin{bmatrix} |\alpha|^2 \mathrm{Tr}_E(|E_+(t)\rangle\langle E_+(t)|) & \alpha\beta^* \mathrm{Tr}_E(|E_+(t)\rangle\langle E_-(t)|) \\ \beta\alpha^* \mathrm{Tr}_E(|E_-(t)\rangle\langle E_+(t)|) & |\beta|^2 \mathrm{Tr}_E(|E_-(t)\rangle\langle E_-(t)|) \end{bmatrix}$$

- Let $|e_i\rangle_E$ be basis set for environment: $\langle e_i|e_j\rangle = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$
  - Then, using the identity $\sum_i |e_i\rangle\langle e_i| = I$,

$$\mathrm{Tr}_E(|E_a(t)\rangle\langle E_b(t)|) = \sum_i \langle e_i|E_a(t)\rangle\langle E_b(t)|e_i\rangle = \sum_i \langle E_b(t)|e_i\rangle\langle e_i|E_a(t)\rangle = \langle E_b(t)| \left(\sum_i |e_i\rangle\langle e_i| \right) |E_a(t)\rangle$$

$$= \langle E_b(t)|E_a(t)\rangle$$

- The density matrix becomes

$$\hat{\rho}_S(t) = \begin{bmatrix} |\alpha|^2 & \alpha\beta^* \langle E_-(t)|E_+(t)\rangle \\ \beta\alpha^* \langle E_+(t)|E_-(t)\rangle & |\beta|^2 \end{bmatrix}$$

  - Hence the environment affects the system's coherences.

The Environment

- Environments consist of large degrees of freedom.
  - Can model as a quantum harmonic oscillator or ensemble of qubits.
  - Consider an environment of N qubits, each in $|\phi_i\rangle_E = \frac{|0\rangle_i + |1\rangle_i}{\sqrt{2}}$ state.
  - Full environment state:

$$|\phi\rangle_E = \bigotimes_{i=1}^{N} \frac{|0\rangle_i + |1\rangle_i}{\sqrt{2}}$$

$$= \frac{|0\rangle_1 + |1\rangle_1}{\sqrt{2}} \otimes \frac{|0\rangle_2 + |1\rangle_2}{\sqrt{2}} \dots \frac{|0\rangle_N + |1\rangle_N}{\sqrt{2}}$$

- The system interacts with the environment/bath.

$$\hat{H}_{int} = \hbar \hat{Z} \otimes \hat{B}$$

  - Let us ignore inter-qubit interactions within the bath.
  - Let us take the bath operator of the form: $\hat{B} = \sum_{i=1}^{N} g_i \hat{Z}_i$.
    - $\hat{Z}_i$ is the Pauli Matrix for environment qubit i.
    - g_i are the coupling strengths between environment qubit i and system S.

Environment States

$$|\phi\rangle_E = \frac{|0\rangle_1 + |1\rangle_1}{\sqrt{2}} \otimes \frac{|0\rangle_2 + |1\rangle_2}{\sqrt{2}} \dots \frac{|0\rangle_N + |1\rangle_N}{\sqrt{2}}$$

$$\hat{H}_{int} = \hbar \hat{Z} \otimes \hat{B}, \qquad \hat{B} = \sum_{i=1}^{N} g_i \hat{Z}$$

- Under this interaction, the environment states evolve to

$$\begin{aligned}
|E_{\pm}(t)\rangle &= e^{\mp i \hat{B} t} |\phi\rangle_E \\
&= e^{\mp i \sum_i g_i \hat{Z}_i t} \frac{|0\rangle_1 + |1\rangle_1}{\sqrt{2}} \otimes \frac{|0\rangle_2 + |1\rangle_2}{\sqrt{2}} \dots \frac{|0\rangle_N + |1\rangle_N}{\sqrt{2}} \\
&= e^{\mp i g_1 \hat{Z}_1 t} \frac{|0\rangle_1 + |1\rangle_1}{\sqrt{2}} \otimes e^{\mp i g_2 \hat{Z}_2 t} \frac{|0\rangle_2 + |1\rangle_2}{\sqrt{2}} \dots e^{\mp i g_N \hat{Z}_N t} \frac{|0\rangle_N + |1\rangle_N}{\sqrt{2}} \\
&= \frac{e^{\mp i g_1 t}|0\rangle_1 + e^{\pm i g_1 t}|1\rangle_1}{\sqrt{2}} \otimes \frac{e^{\mp i g_2 t}|0\rangle_2 + e^{\pm i g_2 t}|1\rangle_2}{\sqrt{2}} \dots \frac{e^{\mp i g_N t}|0\rangle_N + e^{\pm i g_N t}|1\rangle_N}{\sqrt{2}}
\end{aligned}$$

- Now we can calculate coherences in system density matrix

$$\hat{\rho}_S(t) = \begin{bmatrix} |\alpha|^2 & \alpha\beta^* \langle E_-(t)|E_+(t)\rangle \\ \beta\alpha^* \langle E_+(t)|E_-(t)\rangle & |\beta|^2 \end{bmatrix}$$

Coherence Factor

$$\hat{\rho}_S(t) = \begin{bmatrix} |\alpha|^2 & \alpha\beta^*\langle E_-(t)|E_+(t)\rangle \\ \beta\alpha^*\langle E_+(t)|E_-(t)\rangle & |\beta|^2 \end{bmatrix}$$

- The coherence factor is

$$\begin{aligned}
&\langle E_+(t)|E_-(t)\rangle \\
&= \frac{e^{ig_1t}\langle 0|_1 + e^{-ig_1t}\langle 1|_1}{\sqrt{2}} \frac{e^{ig_1t}|0\rangle_1 + e^{-ig_1t}|1\rangle_1}{\sqrt{2}} \\
&\otimes \frac{e^{ig_2t}\langle 0|_2 + e^{-ig_2t}\langle 1|_2}{\sqrt{2}} \frac{e^{ig_2t}|0\rangle_2 + e^{-ig_2t}|1\rangle_2}{\sqrt{2}} \cdots \\
&\cdot \frac{e^{ig_Nt}\langle 0|_N + e^{-ig_Nt}\langle 1|_N}{\sqrt{2}} \frac{e^{ig_Nt}|0\rangle_N + e^{-ig_Nt}|1\rangle_N}{\sqrt{2}} \\
&= \frac{e^{2ig_1t} + e^{-2ig_1t}}{2} \frac{e^{2ig_2t} + e^{-2ig_2t}}{2} \cdots \frac{e^{2ig_Nt} + e^{-2ig_Nt}}{2} \\
&= \cos(2g_1t)\cos(2g_2t)\cdots\cos(2g_Nt) = \Pi_{i=1}^{N}\cos(2g_it)
\end{aligned}$$

Qubit vs. Ensemble Environment

- Single qubit environment ($N = 1$):

$$\langle E_+(t)|E_-(t)\rangle = \cos(2g_1 t)$$

  - Density matrix:

$$\hat{\rho}_S(t) = \begin{bmatrix} |\alpha|^2 & \alpha\beta^* \cos(2g_1 t) \\ \beta\alpha^* \cos(2g_1 t) & |\beta|^2 \end{bmatrix}$$

  - The coherences revive periodically.

- Large environment ($N \to \infty$):

$$\langle E_+(t)|E_-(t)\rangle = \Pi_{i=1}^{N} \cos(2g_i t) \to e^{-\Gamma^2 t^2}$$

  - Density matrix:

$$\hat{\rho}_S(t) \to \begin{bmatrix} |\alpha|^2 & \alpha\beta^* e^{-\Gamma^2 t^2} \\ \beta\alpha^* e^{-\Gamma^2 t^2} & |\beta|^2 \end{bmatrix}$$

  - The coherences decay over time → decoherence.
  - The qubit's quantum information is distributed across N degrees of freedom and is practically inaccessible.

Summary

- We started with a qubit

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

- It interacted with an environment composed of infinite qubits

$$|\phi\rangle_E = \frac{|0\rangle_1 + |1\rangle_1}{\sqrt{2}} \otimes \frac{|0\rangle_2 + |1\rangle_2}{\sqrt{2}} \dots \frac{|0\rangle_N + |1\rangle_N}{\sqrt{2}}$$

- The interaction was of the form

$$\hat{H}_{int} = \hbar\hat{Z} \otimes \hat{B} = \hbar\hat{Z} \otimes \sum_{i=1}^{N} g_i\hat{Z}$$

- Under this interaction, the qubit got entangled with the bath and was no longer in a pure state.

$$|\Psi(t)\rangle_{S+E} = \alpha|0\rangle_S \otimes |E_+(t)\rangle + \beta|1\rangle_S \otimes |E_-(t)\rangle$$

- The system's density matrix became

$$\lim_{N\to\infty} \hat{\rho}_S(t) = \begin{bmatrix} |\alpha|^2 & \alpha\beta^* e^{-\Gamma^2 t^2} \\ \beta\alpha^* e^{-\Gamma^2 t^2} & |\beta|^2 \end{bmatrix}$$

- The environment causes the coherences of the qubit to decay, leading to loss of phase information

$$\lim_{t\to\infty} \hat{\rho}_S(t) = \begin{bmatrix} |\alpha|^2 & 0 \\ 0 & |\beta|^2 \end{bmatrix} = |\alpha|^2|0\rangle\langle 0| + |\beta|^2|1\rangle\langle 1|$$

- Hence, decoherence causes the qubit's state to become a classical mixture.

Measurement Problem

- We can model measurements using this approach.
- Consider
  - Qubit system S with basis states $\{|s_i\rangle|\ i = 1,2\}$.
  - Qubit detection apparatus A with basis states $\{|a_i\rangle|i = 1,2\}$. This plays role of E.
  - Suppose the system-apparatus interaction is $\hat{H}_{int} = \hbar\hat{Z} \otimes \hat{B}$.
- Ideal measurement: apparatus does not influence system during measurement.
  - After interaction, its state corresponds to system's state.

$$|s_i\rangle|a_j\rangle \rightarrow |s_i\rangle|a_i\rangle$$

  - $|a_j\rangle$ is the apparatus state before measurement.
  - The system imprints signal on the apparatus.
- In a superposition $|\psi\rangle_S = \frac{|0\rangle_S + |1\rangle_S}{\sqrt{2}}$
  - The final state is entangled

$$|\Psi\rangle_{S+A} = |\psi\rangle_S \otimes |a_j\rangle = \frac{|0\rangle_S + |1\rangle_S}{\sqrt{2}} \otimes |a_j\rangle = \frac{|0\rangle_S \otimes |0\rangle_A + |1\rangle_S \otimes |1\rangle_A}{\sqrt{2}} = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

  - By linearity of quantum mechanics, the rule applies to each term separately.

Problem 1: The Problem of Preferred Basis

- Given that system $|\psi\rangle$ can be in any basis, how does measurement fix a basis?
- System-Apparatus interaction produces entangled state

$$|\Psi\rangle_{S+A} = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

  - Recall: Entangled states are entangled in all possible basis.
    - Can equivalently write state in x basis

$$|\Psi\rangle_{S+A} = \frac{|+\,+\rangle + |-\,-\rangle}{\sqrt{2}}$$

- The system-apparatus interaction $\hat{H}_{int} = \hbar\hat{Z} \otimes \hat{B}$ selects the eigenstates of $\hat{Z}$, namely $|0\rangle, |1\rangle$.
  - $|0\rangle, |1\rangle$ are the states that are stable under the interaction.
    - These states only pick up phases under interaction $\hat{H}_{int}$.
    - They do not spread into new superpositions.
  - This is called environment-induced Superselection (einselection).
- Decoherence solves the problem of preferred basis. ✓

Problem 2: Non-Observability of Interference

- Why are quantum superpositions not routinely observed macroscopically?
- For interaction of the type $\hat{H}_{int} = \hbar \hat{Z} \otimes \hat{B}$,
  - If system starts off in $|0\rangle_S$ it ends up in $|0\rangle_S \otimes |E_+(t)\rangle$
  - If system starts off in $|1\rangle_S$ it ends up in $|1\rangle_S \otimes |E_-(t)\rangle$
  - If system starts off in $\alpha|0\rangle_S + \beta|1\rangle_S$ it ends up in $\alpha|0\rangle_S \otimes |E_+(t)\rangle + \beta|1\rangle_S \otimes |E_-(t)\rangle$
- Since $\langle E_+(t)|E_-(t)\rangle \to 0$ due to decoherence, the apparatus distinguishes which branch occurred and interference becomes inaccessible.
  - System density matrix:

$$\lim_{t\to\infty} \hat{\rho}_S(t) = \begin{bmatrix} 1/2 & 0 \\ 0 & 1/2 \end{bmatrix} = \frac{|0\rangle\langle 0| + |1\rangle\langle 1|}{2}$$

  - Classical behavior emerges from reduced density matrix, which is a classical mixture with no quantum intereference.
- Decoherence solves the problem of non-observability of interference. ✓

Problem 3: The Problem of Outcomes

- Why does measurement give either $|0\rangle_S$ or $|1\rangle_S$, not both?
- Decoherence does not answer this question. ✗
  - The system state $|\psi\rangle_S = \mathrm{Tr}_A[\hat{\rho}_{S+A}(t)]$ is a classical mixture.
  - But the total state $S + A$ remains pure

$$|\Psi\rangle_{S+A} = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

  - Decoherence stops the interference, but it does not make one branch disappear.
- This is an open problem.
  - Resolution requires choosing an interpretation of quantum mechanics.
    - Copenhagen interpretation
    - Many worlds interpretation
    - …